Technical Report: Airfoil Selection and Aerodynamic Analysis (Problem 1)

1) Candidate Shortlists & Final Selections

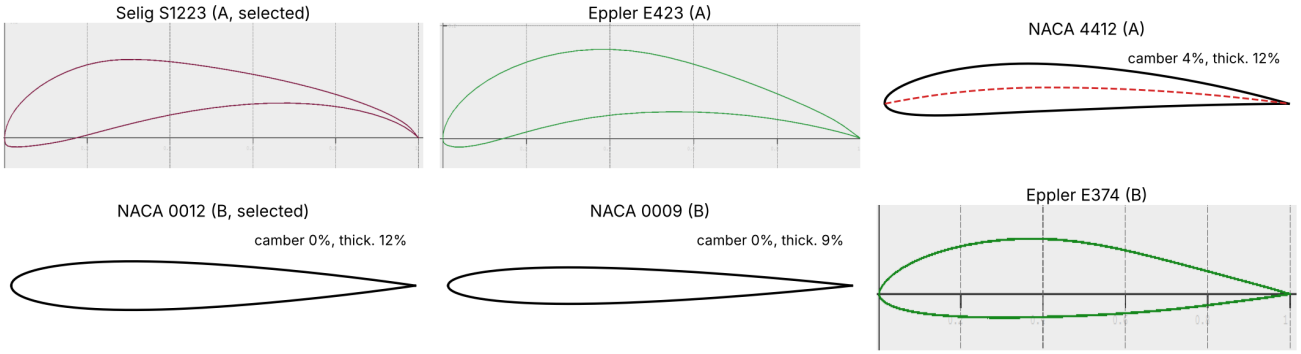
Assumptions. Sea-level air ($\rho = 1.225$ kg/m^3, $\mu = 1.81 \times 10^{-5}$ Pa s). A: $W = 4$ kg, $S = 0.70$ m^2, chord ≈ 0.25 m, speed 8–10 m/s, so $Re = \rho V c/\mu \approx 1.3$–$1.7 \times 10^5$. B: $W = 1.5$ kg, $S = 0.35$ m^2, chord ≈ 0.20 m, speed 15–20 m/s, so $Re \approx 2$–3×10^5. All airfoils were compared at one common $Re = 160{,}000$ for fairness (B flies a little higher, where thin sections do better, so the ranking is conservative).

Aircraft A: Heavy Payload Carrier

- **Objective:** Maximize low-speed lift so the plane can carry the most cargo and still take off in a short distance, while staying stable in flight.
- **Shortlist:** Selig S1223 (high camber), Eppler E423 (high lift), NACA 4412 (baseline).
- **Selected: Selig S1223.**
- **Reasoning:** Its 8.1% camber and concave lower surface give the highest $C_{l,\max}$ of the three (about 2.1–2.2 at $Re = 160{,}000$, versus 1.85 for E423 and 1.40 for NACA 4412). With $W/S = 56$ N/m^2 and a finite-wing $C_{L,\max} \approx 0.8\, C_{l,\max}$, $V_s = \sqrt{2W/(\rho S C_{L,\max})} \approx 7.3$ m/s for S1223 against about 9.1 m/s for NACA 4412. The cost is a strong nose-down moment ($C_m \approx -0.25$), which needs a larger tail download and some trim drag, so the tail must be sized for it to keep the flight stable.

Aircraft B: Highly Maneuverable / Aerobatic

- **Objective:** Same performance upright and inverted, quick roll response and no pitch drift.
- **Shortlist:** NACA 0012 (symmetrical), NACA 0009 (thin symmetrical), Eppler E374 (semi-symmetrical).
- **Selected: NACA 0012.**
- **Reasoning:** Zero camber gives the same lift curve upright and inverted. $C_m \approx 0$ removes trim changes during aggressive manoeuvres. NACA 0009 stalls abruptly (its polar drops from $C_l \approx 0.9$ to 0.5), and E374's camber makes inverted flight less efficient.



2. Aerodynamic Performance Comparison

XFLR5/XFOIL batch analysis at $Re = 160{,}000$ (Geometry = camber % / thickness %)

Airfoil	Role	Geometry	$C_{l,\max}$	$C_{d,\min}$	$(C_l/C_d)_{\max}$	C_m	Stall / Verdict
S1223	A	8.1% / 12.1%	2.1–2.2	0.025	≈50	≈ -0.25	Gradual; **Selected**
E423	A	10.2% / 12.5%	1.85	0.022	≈45	≈ -0.20	Gradual, noisy; runner-up
NACA 4412	A	4.0% / 12.0%	1.40	0.010	≈50	≈ -0.09	Moderate; baseline
NACA 0012	B	0% / 12.0%	1.10	0.008	≈35	0.00	Gentle; **Selected**
NACA 0009	B	0% / 9.0%	0.95	0.007	≈35	0.00	Abrupt; too sharp
E374	B	2.2% / 10.9%	1.40	0.009	≈50	≈ -0.05	Gradual; compromise

$(C_l/C_d)_{\max}$ values are approximate readings from the C_l–C_d polar (±10) and stall types are read from the shape of the C_l–α curve after the peak. Low-Re XFOIL results are noisy near stall (laminar separation bubbles), so trends matter more than the last digit.

3. Geometry Effects & Single-Airfoil Trade-Offs

- **Camber Effects:** Camber shifts the lift curve up ($\alpha_{L=0} < 0$) and raises $C_{l,\max}$, which lowers stall speed, but it makes C_m more negative (S1223: about -0.25), so the tail has to give a larger download. Symmetrical NACA 0012 keeps $C_m \approx 0$.
- **Thickness Effects (12%):** Gives depth for spars and internal space, and a rounder leading edge keeps the flow attached to a higher angle of attack. Thicker sections have higher $C_{d,\min}$ (0.008 for the 0012 against 0.007 for the 0009), but very thin sections stall abruptly at low Re.
- **Leading-edge radius, camber position and Re:** These control the stall type and the laminar bubble. Below about 2×10^5 the bubble adds drag and makes polars noisy.

- **Can one airfoil do both missions?** Not optimally. A wants high camber and $C_{l,\max}$ and accepts a big C_m and drag; B wants zero camber, $C_m \approx 0$ and a gentle stall and accepts $C_{l,\max} \approx 1.1$. S1223 on B would give a large inverted-flight penalty, big trim changes and high drag at speed. NACA 0012 on A would raise V_s from about 7.3 to about 10 m/s, which lengthens the take-off run and cuts the payload. A compromise like E374 ($C_{l,\max} \approx 1.4$, $C_m \approx -0.05$) is second best for both. Flaps can recover some lift for take-off but add weight and complexity.

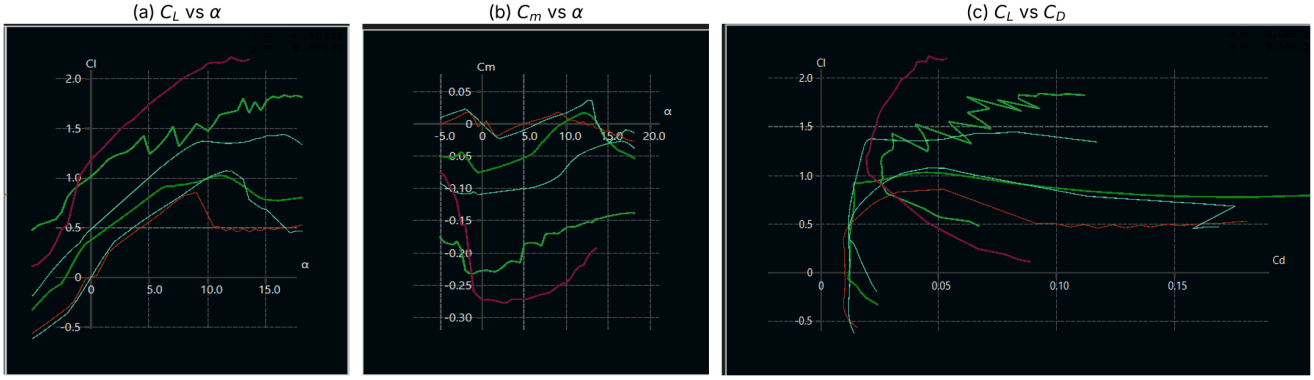
4. 2D Airfoil vs. 3D Whole-Aircraft Factors

- **Airfoil (2D):** $C_{l,\max}$, C_d, C_m, stall behaviour and thickness for structure.
- **3D Induced Drag:** $C_{D,i} = C_L^2/(\pi e AR)$. A high aspect ratio ($AR$) on Aircraft A reduces tip-vortex drag at high C_L; B uses a lower AR for roll rate and stiffness. 3D $C_{L,\max}$ is lower than the 2D value.
- **Wing Loading (W/S):** $V_s = \sqrt{2(W/S)/(\rho C_{L,\max})}$ depends on wing area as much as on the section. A: 56 N/m^2; B: about 42 N/m^2, which helps tight turns.
- **Controls & Thrust:** Manoeuvrability comes from control-surface size and deflection and tail volume, not the airfoil; B needs $T/W 1.0$, A about 0.4–0.5 (typical assumed values).
- **Structure:** Weight and wing stiffness set the payload fraction and the g-limit; a thicker section helps A.

5. Computational Polars & Setup

- **Setup:** XFLR5 v6.62 (XFOIL viscous engine), $Re = 160{,}000$, $M = 0.00$, free transition ($N_{crit} = 9.0$), 120 panel nodes, same settings for all six airfoils.
- **Plot Identification Note:** magenta = S1223 (top curve in C_L–α, about 2.1–2.2), thin green = E423, pale cyan = NACA 4412, cyan = NACA 0012, red = NACA 0009, thick green = E374. NACA 0009 shows the abrupt stall; S1223 shows the most nose-down C_m.
- **Limitations:** The jagged parts near stall come from unconverged points and laminar bubbles at low Re; XFOIL is 2D and post-stall values are unreliable.

(a) C_L vs α (b) C_m vs α (c) C_L vs C_D



References

[1] Selig, M.S. et al., *Summary of Low-Speed Airfoil Data, Vol. 1–3*, SoarTech Publications, 1995–1997 (UIUC Low-Speed Airfoil Tests).
[2] Drela, M., "XFOIL: An Analysis and Design System for Low Reynolds Number Airfoils", in *Low Reynolds Number Aerodynamics*, Springer, 1989.
[3] XFLR5 v6.xx documentation and guidelines, `xflr5.tech`.
[4] Abbott, I.H., von Doenhoff, A.E., *Theory of Wing Sections*, Dover, 1959.
[5] Anderson, J.D., *Fundamentals of Aerodynamics*, McGraw-Hill.
[6] Airfoil Tools (`airfoiltools.com`) and the UIUC Airfoil Coordinates Database (`m-selig.ae.illinois.edu/ads`), airfoil data.